

5 The puma, *Puma concolor*, lives in North and South America.

Fig. 5.1 shows a puma.

Fig 5.2 shows the distribution of the puma species.



Fig. 5.1

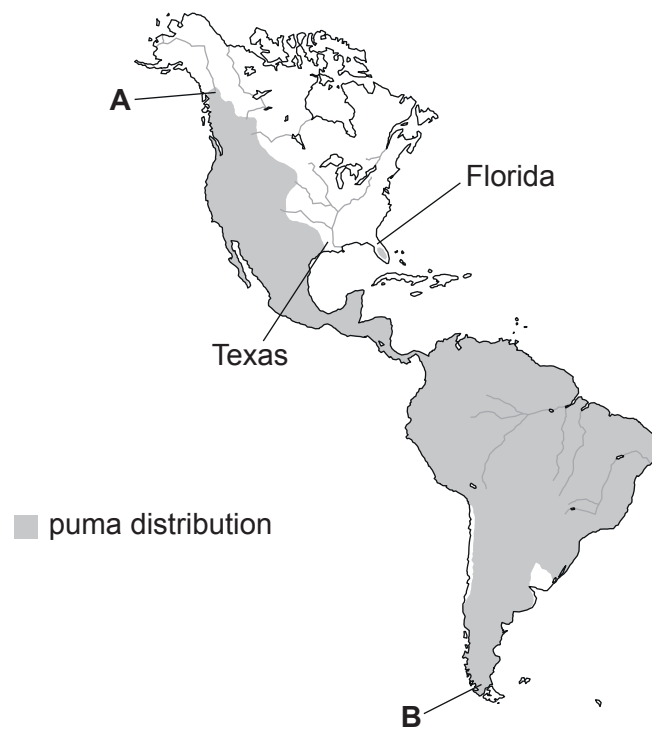


Fig. 5.2

- (a) Members of different subspecies belong to the same species but have some morphological differences and are found in different geographical locations.

In the past the puma has been divided into 32 subspecies. The subspecies of puma varied in body size, coat colour and behaviour to adapt each population to its environment.

Explain how the different subspecies of puma evolved.

[4]

In 2016, genetic analysis concluded that there are only two genetically distinct subspecies of puma, one in North and Central America and one in South America.

- (b)** Outline how practical techniques could be used to conduct a genetic analysis of the puma species.

[3]

- (c) Fig. 5.2 shows the location of an isolated puma population in Florida. In 1990, the size of this population was very small, with fewer than 30 individuals.

Three phenotypic features that vary in pumas are the shape of the tail, the pattern of hair growth on the back and the position of the testes in male pumas.

Variant forms of these phenotypic features that are normally rare occur at a high frequency in the small Florida population. These variant forms are:

- bent tail
- abnormal pattern of hair growth on the back
- testes remain in abdomen (undescended) in some male pumas.

- (i) Predict, with reasons, whether these phenotypic features show a continuous or a discontinuous pattern of variation.

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- (ii) Explain how the small size of the Florida population resulted in a high frequency of these normally rare variant forms.

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- (iii) In 1995, eight puma females from Texas were introduced to Florida to increase the breeding success and future size of the puma population in Florida. In the next 20 years the population grew substantially.

Suggest why the introduced females were taken from Texas and not from points **A** or **B** on Fig. 5.2.

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[Total: 14]